

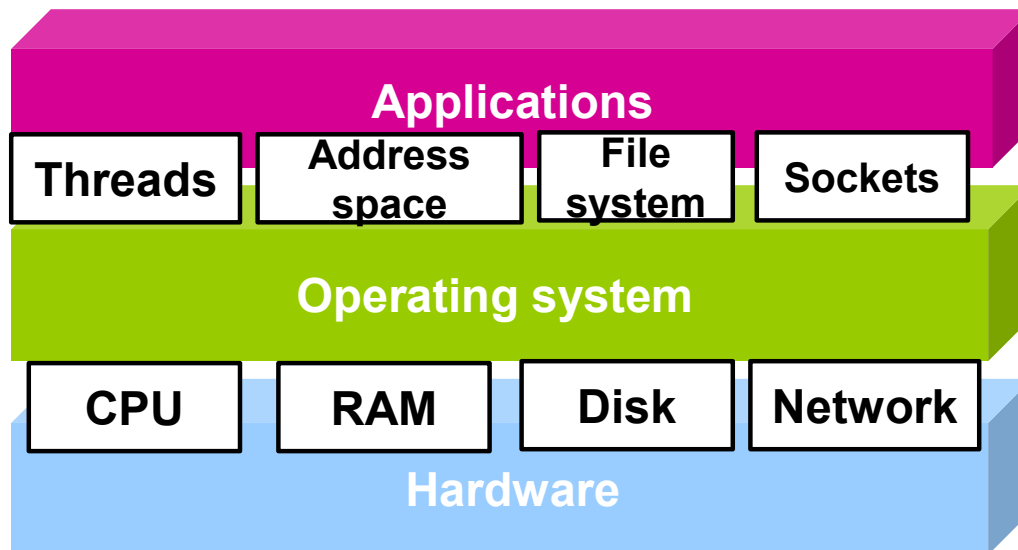
EECS 482

Introduction to ~~operating~~ systems
concepts

Case studies

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~~Operating~~ system roles



- Abstraction
- Management of shared resources

~~Operating~~ system goals

- Convenience
 - Abstraction; removal of resource limits
- Performance
 - Throughput, latency, deadlines
- Fairness
 - Variance, guarantees
- Reliability
 - Correctness in the presence of failure
- Security
 - Correctness in the presence of attacks

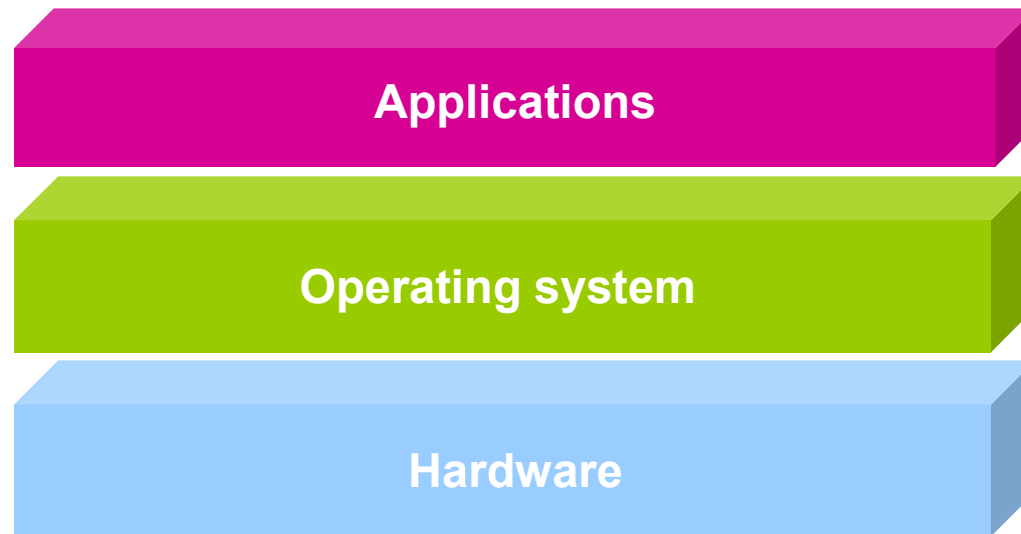
~~Operating~~ system techniques

- Abstraction
- Concurrency
- Atomicity
- Indirection
- Caching
- Prediction
- Persistence
- Naming and translation
- Scheduling/ordering
- Deferring (or anticipating) work
- Resource trading

Case study

- Virtual machines (e.g., VMware)
 - Software layer below the operating system

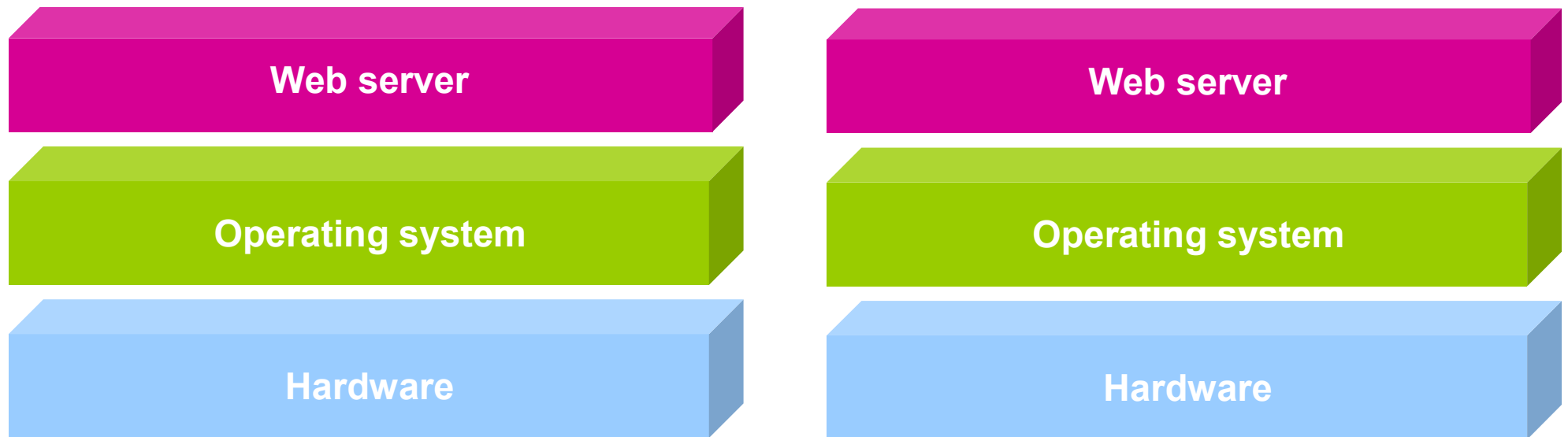
Virtual machines



- Operating systems manage the computer and provide abstractions and services for application programs
- But who manages the computer and provides abstractions and services for the operating system?

Utilize computing resources efficiently and securely

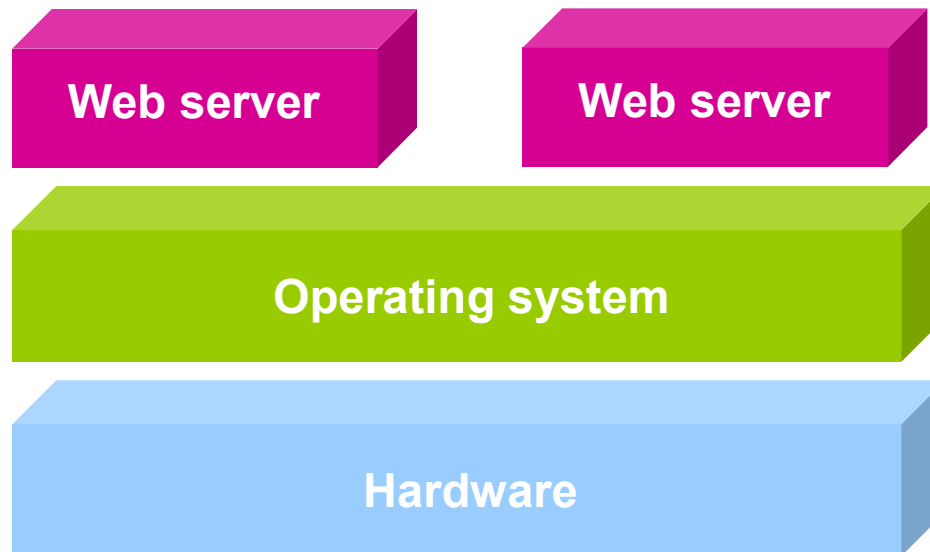
- Option 1: Each web server runs on dedicated machine



- Problems?

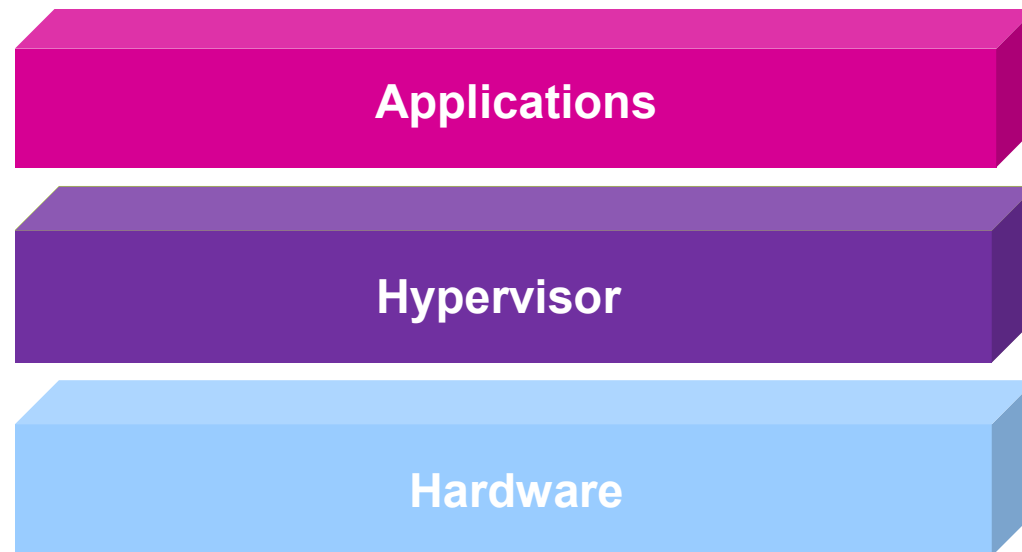
Utilize computing resources efficiently and securely

- Option 2: Multiple web servers share one machine



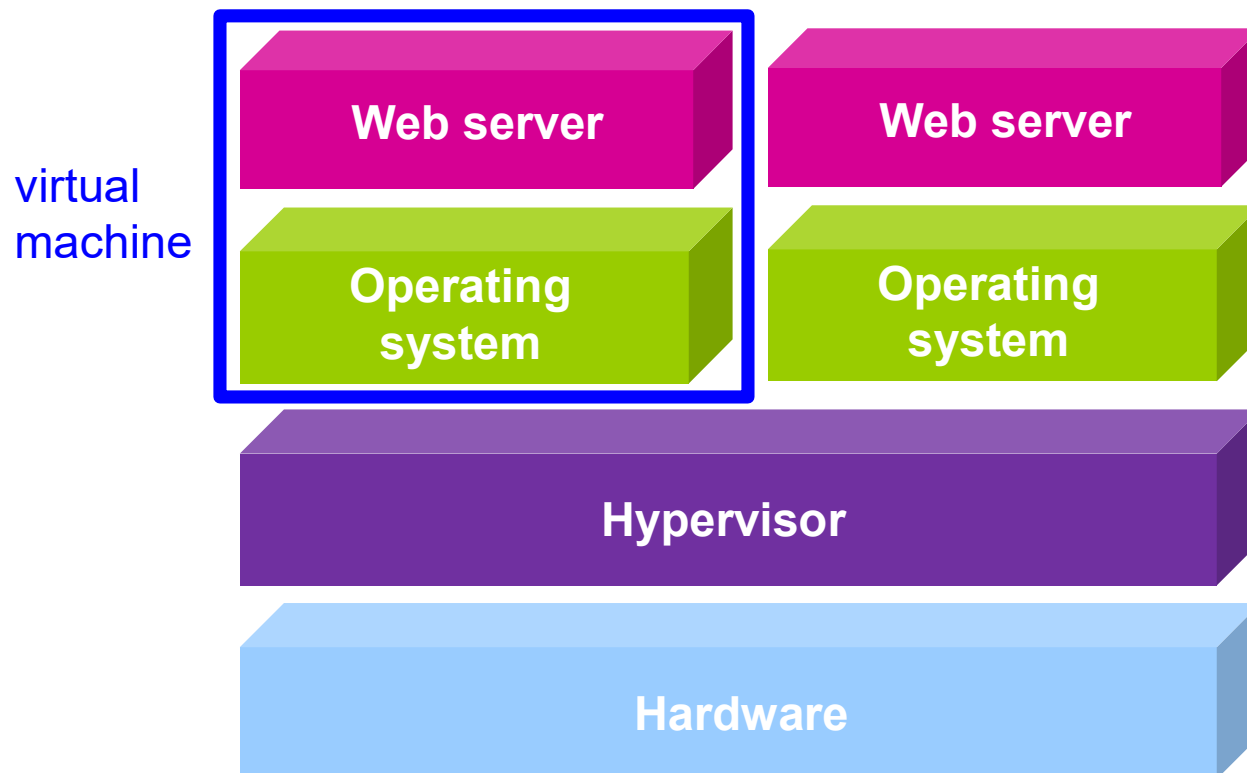
- Problems?

System structure with virtual machines

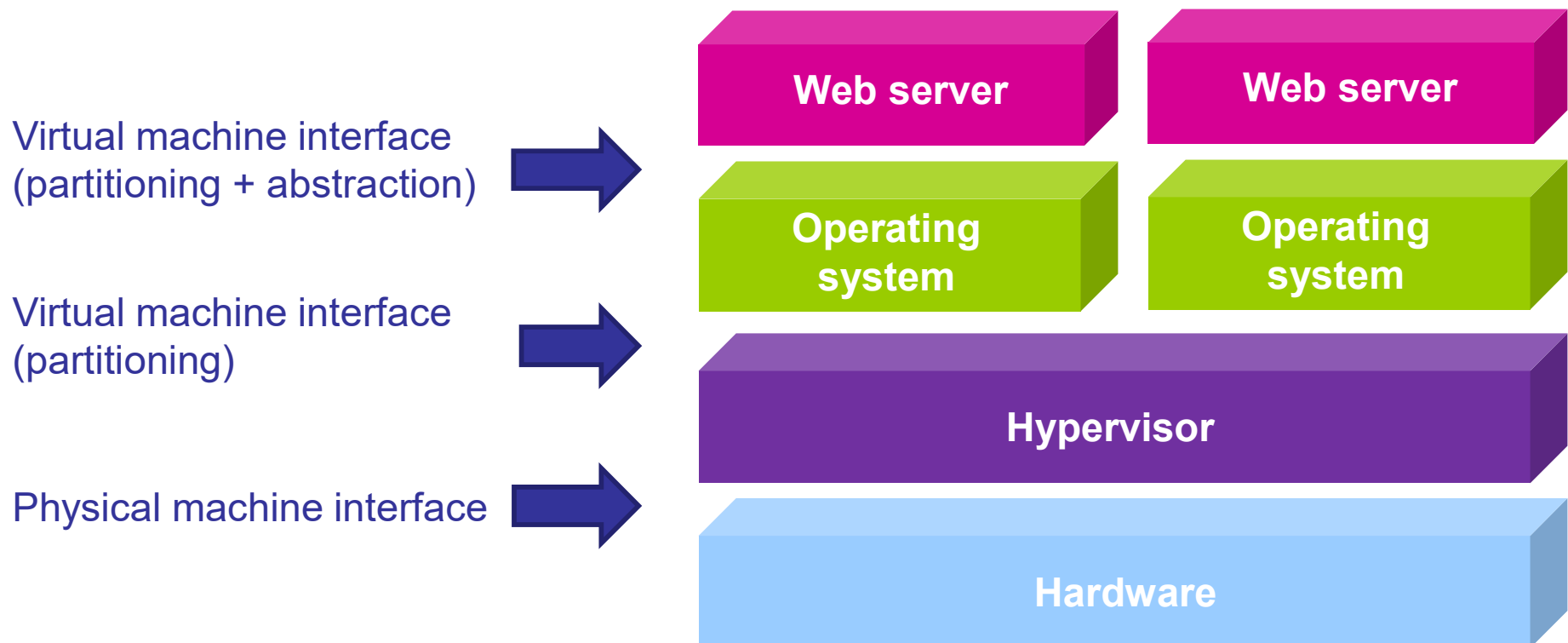


Utilize computing resources efficiently and securely

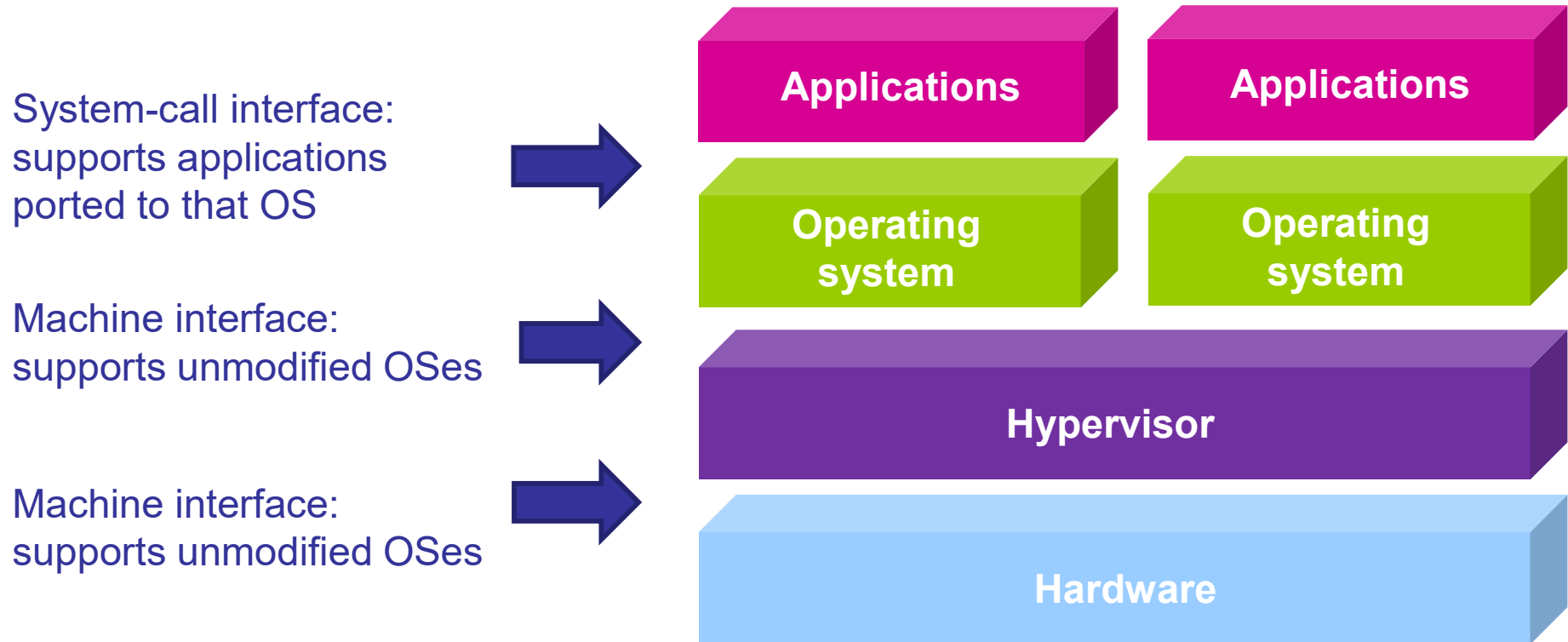
- Option 3: Each web server runs on dedicated virtual machine; multiple virtual machines share a physical machine



Why is this more secure?



Why is this hard?



- Techniques
 - Binary translation
 - Composition of multiple translation levels
 - Hardware virtualization support

What services/abstraction/management could you provide for an operating system?

- Server consolidation
- Simulate different hardware
- Malware analysis, anti-virus
- Fault tolerance
- VM migration
- Debugging OS

Some open questions

- Do we need both hypervisors and operating systems?
- Is there a more efficient way to provide isolation?
- What happens if you add too much functionality to hypervisor?

I hope you have...

- Learned OS concepts and techniques
- Become better at designing, writing, testing, and debugging complex systems
- Enjoyed the class
- Best wishes on the final and the rest of your career!